

Step1a–Step1b: Diagnostic and Generalization Study of the Perturbed Fenchel–Young Surrogate

Weikang

May 2026

1 Purpose of Step1a

Step1a is an in-sample diagnostic experiment designed to visualize how two training objectives move a two-dimensional reward model through the decision landscape of a synthetic kidney exchange problem. It is not intended to prove held-out generalization. Its purpose is to answer the following diagnostic question:

On a fixed batch of synthetic KEP instances, do MSE reward fitting and decision-focused Fenchel–Young (FY) training move through different regions of the two-dimensional synthetic-label decision-gap landscape?

The experiment is useful because it makes the FY surrogate visually inspectable: we can plot both training trajectories in the same parameter plane and evaluate each point by the downstream KEP objective gap under the synthetic label weights.

Important language. Throughout this report, we use *synthetic-label decision gap* or *oracle objective gap*. We avoid the phrase “true regret”, because the evaluation weights are synthetic labels rather than real clinical outcomes.

2 Dataset and Prediction Model

The current Step1a diagnostic uses the controlled noisy-linear synthetic reward dataset:

`dataset/processed/step1_noisy_linear_sigma010.`

Each edge e has a synthetic label weight

$$w_e^{\text{syn}} = \max\{0, (10u_e + 5c_e)(1 + \xi_e)\},$$

where u_e is the edge utility, c_e is the recipient cPRA feature, and the multiplicative noise scale is $\sigma = 0.10$.

The learned reward model is the two-feature linear probe

$$\hat{w}_e(\theta) = \theta_1 u_e + \theta_2 c_e,$$

with

$$\theta_1 \leftrightarrow \text{utility}, \quad \theta_2 \leftrightarrow \text{recipient cPRA}.$$

The vector $[10, 5]$ is the *noise-free linear coefficient*; it should not be called the true parameter, because the labels include multiplicative edge-level noise.

3 Training Objectives

3.1 MSE reward-fitting baseline

The two-stage baseline trains θ by reward fitting:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \frac{1}{|E_i|} \sum_{e \in E_i} (\hat{w}_e^i(\theta) - w_e^{i,\text{syn}})^2.$$

This objective tries to recover the synthetic reward labels. In this controlled setting, it is expected to move toward the clean signal direction [10, 5].

3.2 Decision-focused FY training

FY training updates the same two-dimensional model using the perturbed optimizer surrogate. For a graph instance, let

$$y^* = S(w^{\text{syn}})$$

be the KEP solution optimized under the synthetic label weights. The FY objective used in this diagnostic is, up to constants independent of θ ,

$$L_{\text{FY}}(\theta) \approx \mathbb{E}_z \left[\max_{y \in \mathcal{Y}} \langle \hat{w}(\theta) + z, y \rangle \right] - \langle \hat{w}(\theta), y^* \rangle.$$

The corresponding gradient estimate has the form

$$\nabla_{\theta} L_{\text{FY}}(\theta) \approx X^{\top} (\bar{y}_{\epsilon}(\hat{w}(\theta)) - y^*),$$

where $\bar{y}_{\epsilon}$ is the average KEP solution under perturbed predicted weights.

4 Evaluation Metric

For each parameter vector θ , Step1a evaluates the downstream decision quality by the synthetic-label decision gap:

$$G(\theta) = \frac{1}{N} \sum_{i=1}^N \left[(w^{i,\text{syn}})^{\top} y_i^* - (w^{i,\text{syn}})^{\top} S(\hat{w}^i(\theta)) \right].$$

Here $y_i^* = S(w^{i,\text{syn}})$ is the oracle KEP solution under the synthetic label weights, and $S(\hat{w}^i(\theta))$ is the KEP solution induced by the predicted weights. Smaller $G(\theta)$ is better.

5 Experimental Configuration

The historical completed Step1a diagnostic used the following configuration.

6 Results: Summary Table

The table suggests three diagnostic observations. First, the MSE baseline recovers the noise-free linear coefficient direction, but does not reach zero decision gap because the two-feature probe cannot represent the deterministic multiplicative edge noise. Second, FY with $\epsilon = 0.2$ visits a lower-gap region than the final MSE endpoint, but the best-gap point occurs early. Third, larger perturbation $\epsilon = 1.0$ changes the scale and timing of the FY trajectory.

Table 1: Step1a diagnostic configuration.

Item	Value
Dataset	dataset/processed/step1_noisy_linear_sigma010
Number of graph instances	100
Training epochs	1000
Seed	42
Initialization	(1.6236, 3.3521)
MSE learning rate	0.1
FY learning rate	0.1
FY perturbation samples M	16
Grid size for landscape	25×25
Epsilon values	0.2, 1.0

Table 2: Historical Step1a results on the noisy-linear diagnostic. The decision gap is the average oracle objective gap under synthetic labels.

Method	ϵ	Final θ	Final FY objective	Final decision gap	Best decision gap
MSE	–	(10.0017, 5.0147)	–	0.8759	0.8759
FY	0.2	(2.8135, 1.3589)	0.3609	0.8714	0.7183 at epoch 42
FY	1.0	(13.9796, 6.7825)	1.8042	0.8714	0.8668 at epoch 781

7 Figure-Based Analysis

7.1 $\epsilon = 0.2$: diagnostic summary

How to read this group. The contour panel shows that MSE reward fitting moves toward the noise-free linear coefficient $[10, 5]$, while FY follows the geometry induced by the KEP solver and the perturbed FY objective. The metric panels separate the training surrogate from the deterministic decision metric: the best decision-gap epoch can occur before the final FY iterate, motivating validation-based checkpoint selection rather than only using the final epoch.

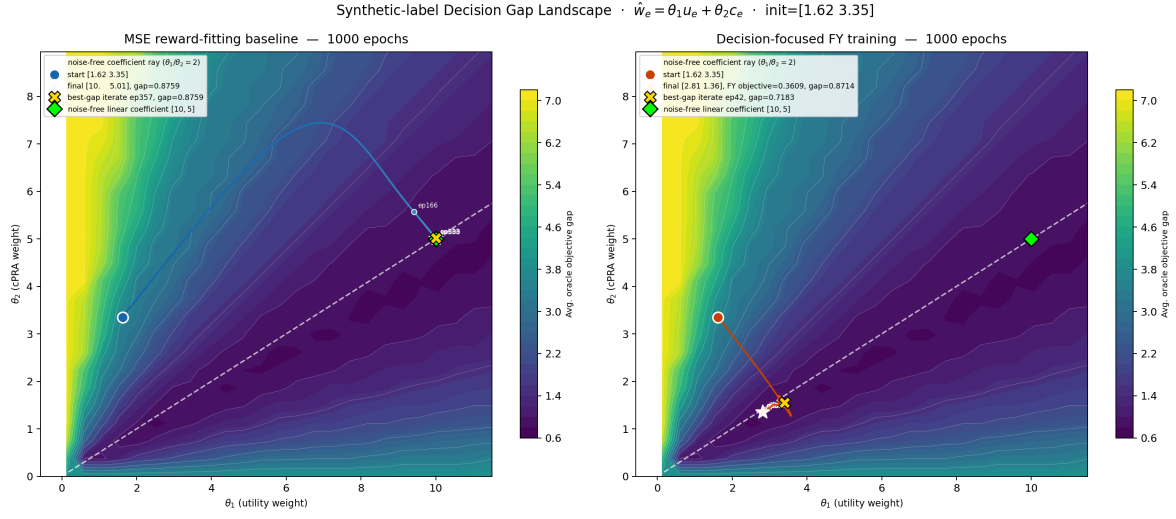
7.2 $\epsilon = 1.0$: diagnostic summary

How to read this group. With a larger perturbation scale, FY tends to move to a larger parameter scale. This is consistent with the fact that the model must produce decisions that are stable under stronger perturbations. The 3D and epoch views help distinguish movement in parameter space from movement in the decision metric and the FY surrogate objective.

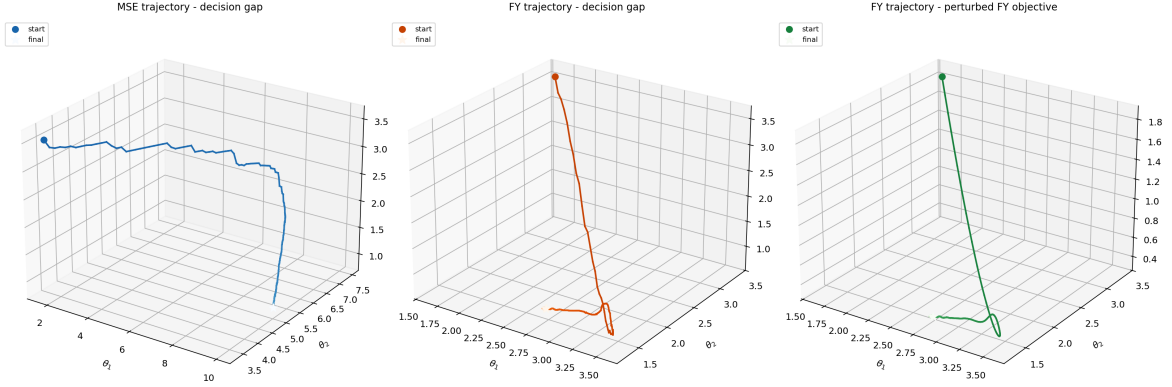
8 Interpretation

Step1a supports the following qualitative conclusions.

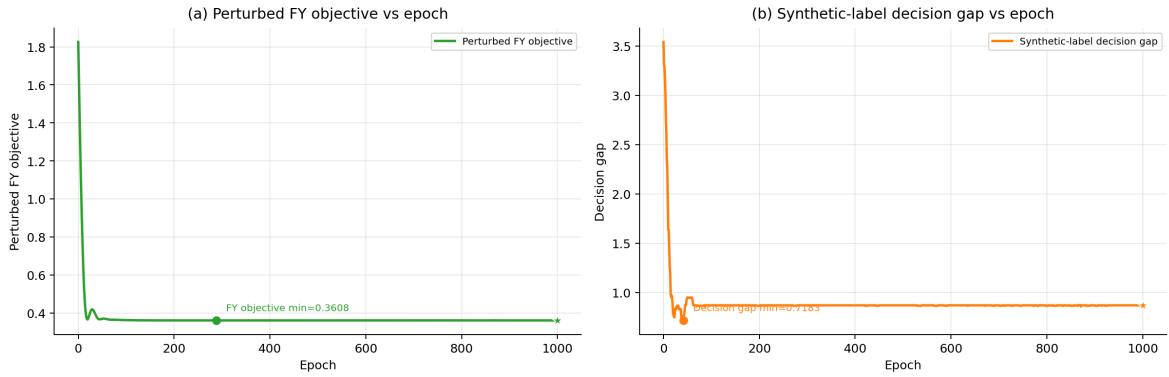
1. **MSE reward fitting follows the reward-label geometry.** It moves toward the noise-free linear coefficient $[10, 5]$, because its objective is to fit edge weights.
2. **FY training follows a decision-induced geometry.** It updates parameters according to differences between perturbed optimizer outputs and the synthetic-label oracle KEP solution.



(a) MSE and FY trajectories on the synthetic-label decision-gap landscape.

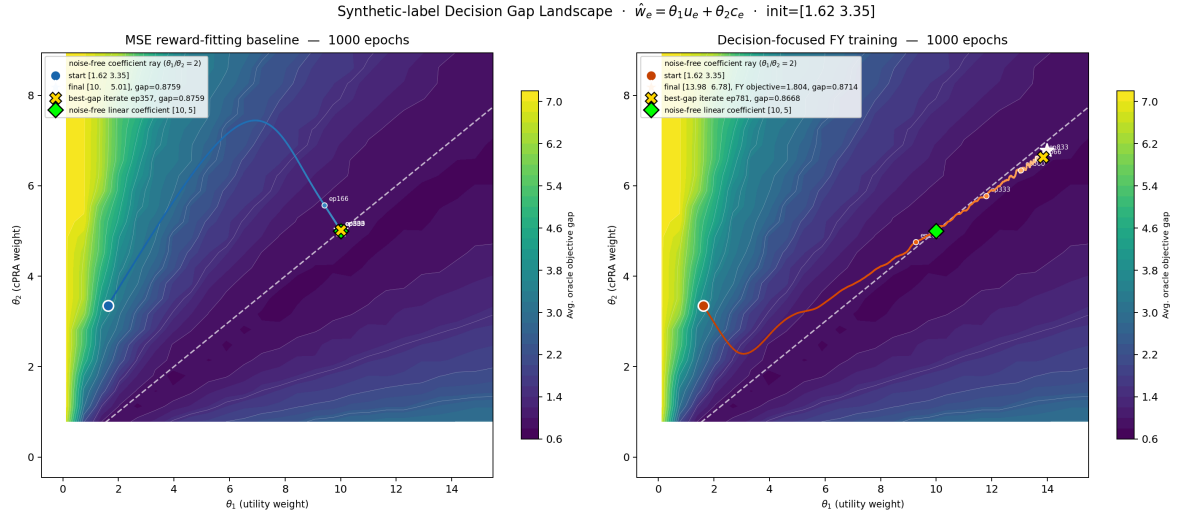


(b) Three trajectory views: MSE decision gap, FY decision gap, and FY perturbed objective.
Step 1a FY diagnostics over epochs, epsilon=0.2

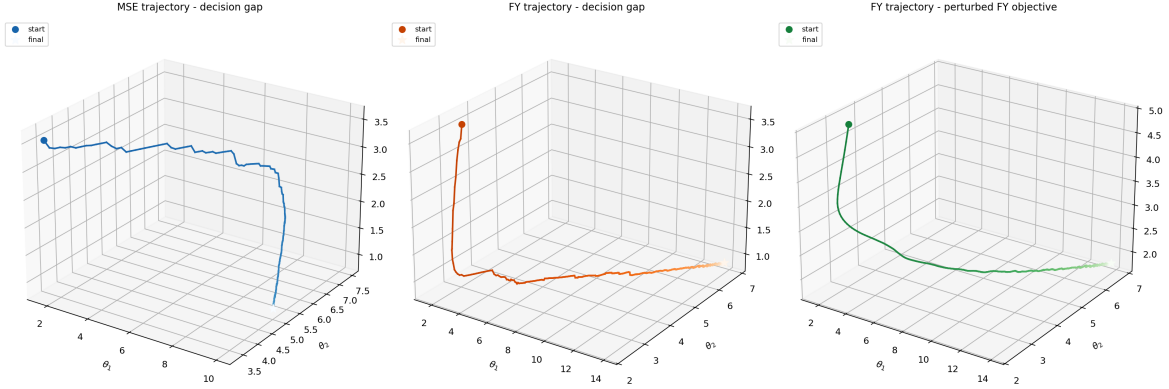


(c) FY perturbed objective and synthetic-label decision gap over epochs.

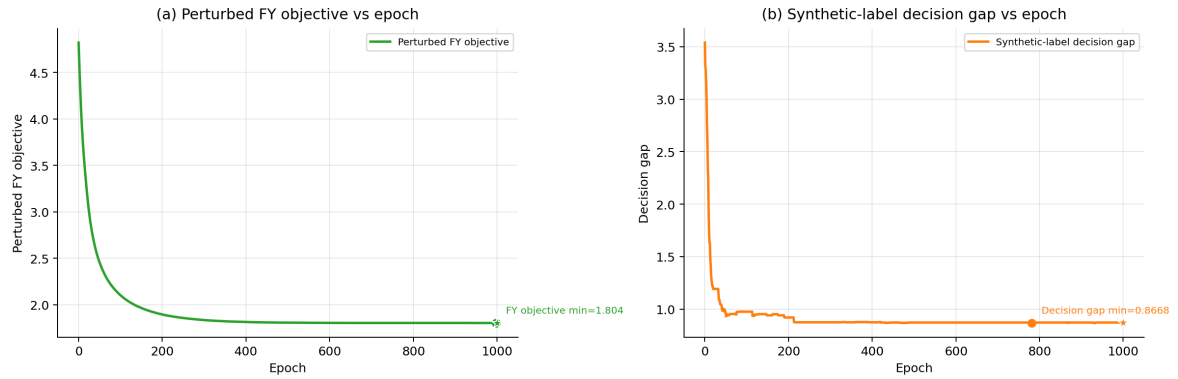
Figure 1: $\epsilon = 0.2$. Step1a in-sample diagnostic summary. The contour plot shows that FY visits a low decision-gap region early; the epoch plot highlights why validation-based checkpoint selection is useful in Step1b.



(a) MSE and FY trajectories on the synthetic-label decision-gap landscape.



(b) Three trajectory views: MSE decision gap, FY decision gap, and FY perturbed objective.
Step 1a FY diagnostics over epochs, epsilon=1.0



(c) FY perturbed objective and synthetic-label decision gap over epochs.

Figure 2: $\epsilon = 1.0$. Step1a in-sample diagnostic summary. The larger perturbation scale changes the scale and timing of the FY trajectory, so the endpoint is not expected to coincide with the clean coefficient [10, 5].

3. **The best deterministic decision point need not be the final FY endpoint.** For $\epsilon = 0.2$, the best observed decision gap appears early, while the FY objective continues to optimize its own perturbation-smoothed criterion.
4. **The perturbation scale changes the path and parameter scale.** Larger ϵ encourages larger parameter scale and different trajectory timing.
5. **Step1a motivates Step1b.** Because the best decision-gap iterate may occur before the final iterate, and because this experiment is in-sample, we need Step1b to use train/validation/test splitting and validation-based checkpoint selection.

9 What Step1a Does Not Claim

Step1a should not be interpreted as evidence of held-out generalization. It uses the same sampled graph instances for trajectory training, metric evaluation, and decision-gap landscape construction. Therefore, the correct claim is:

The FY surrogate can move the two-feature probe through low synthetic-label decision-gap regions on a fixed controlled batch.

The stronger claim,

FY generalizes better than MSE on unseen KEP instances,

belongs to Step1b.

10 Transition to Step1b

Step1a reveals that the final FY iterate is not necessarily the best point in terms of deterministic decision quality. This naturally motivates Step1b, where we train on different numbers of graph instances, select checkpoints using validation metrics, and evaluate the selected models on held-out test instances.

Step1b: Sample-Size Generalization Study for the Perturbed Fenchel–Young Surrogate

11 Purpose of Step1b

Step1b is the held-out generalization companion to the Step1a landscape diagnostic. Step1a asks an in-sample visualization question: on a fixed batch of synthetic KEP instances, how do MSE and perturbed Fenchel–Young (FY) training trajectories move in the two-dimensional parameter space? Step1b asks the standard ML-style question:

As the number of training graph instances increases, do the learned reward models improve held-out kidney exchange decision quality on validation and test graphs that were not used for training?

The experiment is designed to make this question reproducible through a fixed train/validation/test split, validation-based checkpoint selection, and paired test-set comparisons between the two-stage baseline and decision-focused FY training.

Language. This is a synthetic benchmark. The evaluation labels are synthetic reward weights, not clinical outcomes. Therefore the main metric is called *synthetic-label decision gap* or *oracle objective gap*, not “true regret”.

12 Synthetic Benchmark and Model Class

The default Step1b benchmark is the same controlled noisy-linear label regime as Step1a:

$$w_e^{\text{syn}} = \max\{0, (10u_e + 5c_e)(1 + \xi_e)\},$$

where u_e is the edge utility, c_e is the recipient cPRA feature, and the multiplicative noise scale is $\sigma = 0.10$.

Both methods use the same two-feature linear probe:

$$\hat{w}_e(\theta) = \theta_1 u_e + \theta_2 c_e.$$

The vector $[10, 5]$ is the noise-free linear coefficient, not an exact noise-corrupted label parameter.

13 Data Splitting Design

Step1b uses a fixed master split from

`dataset/processed/step1_noisy_linear_sigma010.`

The default split sizes are shown in Table 3.

For each run, a training subset of size

$$n \in \{50, 200, 600, 1200\}$$

is selected reproducibly from the 1200-graph train pool. With the same subset seed, larger training sizes are selected by reshuffling the same train pool and taking a longer prefix. This makes train-size comparisons stable.

Table 3: Step1b master split design.

Split component	Number of graphs
Train pool	1200
Validation split	400
Held-out test split	400

Current archive status. The current formal archive used in this report contains completed runs for $n = 50, 200, 600$. The $n = 1200$ run is planned and can be added to the same figures/tables once completed.

14 Training Methods

14.1 Two-stage reward-fitting baseline

The two-stage method first trains θ by MSE reward fitting:

$$\min_{\theta} \frac{1}{N} \sum_i \frac{1}{|E_i|} \sum_{e \in E_i} (\hat{w}_e^i(\theta) - w_e^{i,\text{syn}})^2.$$

The checkpoint is selected by validation MSE loss. During evaluation, the KEP solver is run with the learned predicted weights $\hat{w}(\theta)$, and the resulting decision is scored under the synthetic label weights.

14.2 Decision-focused FY training

The e2e model uses the same linear probe but trains with the perturbed FY surrogate. For each training graph, the gradient estimate compares the average perturbed optimizer solution with the synthetic-label oracle solution:

$$\nabla_{\theta} L_{\text{FY}}(\theta) \approx X^{\top} (\bar{y}_{\epsilon}(\hat{w}(\theta)) - y^*),$$

where $y^* = S(w^{\text{syn}})$ and

$$\bar{y}_{\epsilon}(\hat{w}) = \frac{1}{M} \sum_{m=1}^M S(\hat{w} + z^{(m)}).$$

Step1b saves two e2e checkpoints:

1. **e2e selected by validation decision gap:** this asks whether the FY trajectory produces a model with good held-out downstream decision quality.
2. **e2e selected by validation FY loss:** this asks whether the FY surrogate itself can serve as a validation/model-selection loss.

These two checkpoints are conceptually different and should both be reported.

15 Evaluation Metrics

For a graph i , the synthetic-label decision gap is

$$G_i(\theta) = (w^{i,\text{syn}})^\top y_i^\star - (w^{i,\text{syn}})^\top S(\hat{w}^i(\theta)).$$

The reported mean decision gap is $\frac{1}{N} \sum_i G_i(\theta)$. The normalized decision gap is

$$\frac{G_i(\theta)}{|(w^{i,\text{syn}})^\top y_i^\star| + \varepsilon},$$

averaged over graphs.

Step1b also reports the achieved/oracle objective ratio:

$$\frac{(w^{i,\text{syn}})^\top S(\hat{w}^i(\theta))}{(w^{i,\text{syn}})^\top y_i^\star}.$$

For paired e2e-vs-2stage comparison, the per-graph improvement is

$$\Delta_i = G_i(\theta_{2\text{stage}}) - G_i(\theta_{\text{e2e}}).$$

Positive Δ_i means that e2e has a smaller decision gap than 2stage on the same test graph. Step1b reports the paired mean improvement and a bootstrap 95% confidence interval.

16 Formal Run Configuration

The current formal run used the configuration in Table 4.

Table 4: Formal Step1b configuration used for the current archive.

Item	Value
Dataset	dataset/processed/step1_noisy_linear_sigma010
Train sizes completed	50, 200, 600
Train size planned	1200
Split seed / subset seed / theta seed	42 / 42 / 42
2stage epochs	500
e2e epochs	500
FY perturbation scale ϵ	1.0
FY perturbation samples M	16
Metric stride	10
Held-out test size	400
Unseen noisy-linear test size	1000
Realistic-synthetic stress test size	2000

17 Evaluation Sets

Step1b currently evaluates models on three datasets.

The realistic-synthetic stress test should not be described as a fully graph-unseen test, because it may share graph structures with the original raw batch. It is better described as a label-regime or distribution-shift stress test.

Table 5: Evaluation datasets used in the current Step1b report.

Dataset	Size	Purpose
Held-out noisy-linear test	400	Standard test split from the original master split
Unseen noisy-linear test	1000	Newly generated graph instances with same label regime
Realistic-synthetic stress test	2000	Different reward-label regime / distribution-shift stress test

18 Result Tables

18.1 Held-out 400-graph noisy-linear test

Table 6 shows the formal held-out test results from the current archive. This is the standard test split associated with the train/validation/test master split.

Table 6: Held-out 400-graph noisy-linear test results. Positive paired improvement means e2e has a smaller decision gap than 2stage on the same test graphs.

Train size	Method	Selected by	Epoch	θ_1	θ_2	Mean gap	Norm. gap	Paired improvement
50	2stage	val MSE	500	9.8707	5.1130	0.8480	0.00611	–
50	e2e	val gap	50	5.4807	2.8132	0.8220	0.00592	0.0260
50	e2e	val FY	440	11.8304	5.0208	0.7587	0.00547	0.0894
200	2stage	val MSE	500	9.8848	5.1038	0.8451	0.00610	–
200	e2e	val gap	150	8.8078	4.5582	0.8480	0.00611	-0.0029
200	e2e	val FY	500	12.2682	6.0649	0.7730	0.00563	0.0721
600	2stage	val MSE	500	9.8899	5.0963	0.8223	0.00591	–
600	e2e	val gap	110	7.7838	4.0153	0.8223	0.00591	0.0000
600	e2e	val FY	490	12.2328	5.8586	0.7507	0.00550	0.0716

Reading Table 6. On the original 400-graph held-out split, the e2e checkpoint selected by validation FY loss is better than the 2stage baseline for the completed train sizes. The e2e checkpoint selected by validation decision gap is not consistently better; it is slightly better at $n = 50$, worse at $n = 200$, and tied at $n = 600$. This distinction is important: the FY-loss-selected model is the stronger evidence that the FY surrogate can sometimes serve as a useful model-selection objective on the original noisy-linear split.

18.2 1000-graph unseen noisy-linear test

Table 7 evaluates the same trained models on a separately generated unseen 1000-graph noisy-linear dataset.

Reading Table 7. The 1000-graph unseen test does not replicate the stronger FY-loss-selected improvements from the 400-graph held-out split. The 2stage baseline is stable around mean gap 0.731, and e2e is tied or slightly worse except for a small improvement at $n = 200$ selected by validation FY loss. The confidence interval for that improvement still crosses zero. Thus, the unseen evaluation makes the claim more cautious: the current two-feature FY model does not yet show robust out-of-split improvement.

Table 7: 1000-graph unseen noisy-linear test results.

Train size	Method	Selected by	Epoch	θ_1	θ_2	Mean gap	Norm. gap	Paired improvement	95% CI
50	2stage	val MSE	500	9.8707	5.1130	0.7311	0.00531	—	—
50	e2e	val gap	50	5.4807	2.8132	0.7352	0.00534	-0.0041	[-0.0114, 0.0006]
50	e2e	val FY	440	11.8304	5.0208	0.7591	0.00550	-0.0280	[-0.0878, 0.0319]
200	2stage	val MSE	500	9.8848	5.1038	0.7313	0.00532	—	—
200	e2e	val gap	150	8.8078	4.5582	0.7313	0.00532	0.0000	[0.0000, 0.0000]
200	e2e	val FY	500	12.2682	6.0649	0.7262	0.00528	0.0050	[-0.0200, 0.0320]
600	2stage	val MSE	500	9.8899	5.0963	0.7309	0.00531	—	—
600	e2e	val gap	110	7.7838	4.0153	0.7329	0.00533	-0.0019	[-0.0078, 0.0000]
600	e2e	val FY	490	12.2328	5.8586	0.7488	0.00545	-0.0179	[-0.0564, 0.0219]

18.3 Realistic-synthetic stress test

Table 8 evaluates the same noisy-linear-trained models under a more complex realistic-synthetic label regime:

$$\text{expected transplant count} \times \text{QALY} \times \text{priority multiplier} \times (1 + \text{deterministic noise}).$$

Table 8: Realistic-synthetic label-regime stress test.

Train size	Method	Selected by	Epoch	θ_1	θ_2	Mean gap	Norm. gap	Ratio	Paired improvement	95% CI
50	2stage	val MSE	500	9.8707	5.1130	12.1103	0.06603	0.93397	—	—
50	e2e	val gap	50	5.4807	2.8132	12.1314	0.06613	0.93387	-0.0210	[-0.0750, 0.0239]
50	e2e	val FY	440	11.8304	5.0208	12.3439	0.06718	0.93282	-0.2335	[-0.4036, -0.0678]
200	2stage	val MSE	500	9.8848	5.1038	12.0837	0.06588	0.93412	—	—
200	e2e	val gap	150	8.8078	4.5582	12.0879	0.06591	0.93409	-0.0041	[-0.0118, 0.0000]
200	e2e	val FY	500	12.2682	6.0649	12.1739	0.06631	0.93369	-0.0902	[-0.1674, -0.0198]
600	2stage	val MSE	500	9.8899	5.0963	12.1108	0.06606	0.93394	—	—
600	e2e	val gap	110	7.7838	4.0153	12.1108	0.06606	0.93394	0.0000	[0.0000, 0.0000]
600	e2e	val FY	490	12.2328	5.8586	12.1670	0.06618	0.93382	-0.0562	[-0.1618, 0.0471]

Reading Table 8. Under the shifted realistic-synthetic reward labels, the noisy-linear-trained 2stage model is slightly better than or tied with the e2e models. This should not be interpreted as a general failure of FY. It mainly shows that a two-feature model trained on controlled noisy-linear labels does not transfer cleanly to a different, more complex reward definition.

19 Figure Roadmap

The current Step1b plotting script produces 10 presentation-ready figures. Each figure is saved as both PNG and PDF under *Step1b/plot_results*.

20 Figure-by-Figure Presentation Notes

20.1 Main decision quality: normalized decision gap

How to explain this figure. Start with this figure because it is the most compact summary of generalization. The left panel is the original held-out noisy-linear split; the middle panel is the newly generated unseen noisy-linear test; the right panel is the realistic-synthetic stress test.

Table 9: Step1b figure roadmap.

Figure	File	What to explain
1	01_normalized_decision_gap_by_dataset	Main generalization plot. Compare normalized decision gap across train sizes and datasets.
2	02_raw_decision_gap_by_dataset	Raw oracle objective gap. Useful for checking whether normalized trends are driven by scale.
3	03_achieved_oracle_ratio_by_dataset	Achieved/oracle objective ratio. Higher is better and easier to interpret as percentage of oracle value.
4	04_paired_improvement_over_2stage	Most important comparison plot. Positive bars mean e2e beats 2stage on paired test graphs.
5	05_selected_parameter_endpoints	Shows where selected checkpoints land in (θ_1, θ_2) space.
6	06_parameter_trajectories	Shows training dynamics in parameter space for each train size.
7	07_2stage_mse_loss_curves	Checks whether MSE reward fitting converges and generalizes on validation MSE.
8	08_e2e_fy_objective_curves	Checks FY surrogate optimization and train/validation FY behavior.
9	09_e2e_decision_gap_curves	Shows whether e2e trajectory improves downstream decision gap on train/validation.
10	10_e2e_validation_tradeoff	Directly compares validation FY objective vs validation decision gap across epochs.

The key message is that FY-loss selection looks promising on the original held-out split, but the advantage weakens or disappears on the 1000-graph unseen test and does not transfer to the shifted realistic-synthetic reward labels.

20.2 Raw decision gap

How to explain this figure. Use this figure to confirm that the normalized-gap trends are not an artifact of normalization. The raw gaps are much larger in the realistic-synthetic stress test because the reward scale is different.

20.3 Achieved/oracle objective ratio

How to explain this figure. This figure translates the decision gap into an “oracle percentage” view. It is useful for communication: even when raw gaps differ across datasets, the ratio shows how close the induced KEP solution is to the synthetic-label oracle solution.

20.4 Direct method performance

How to explain this figure. This plot avoids the large visual uncertainty of paired-improvement confidence intervals and instead shows the three selected models on the same absolute performance metric. It should be read together with the paired statistics in the tables: lower bars indicate better mean normalized decision quality, while the paired tables still provide graph-by-graph significance evidence.

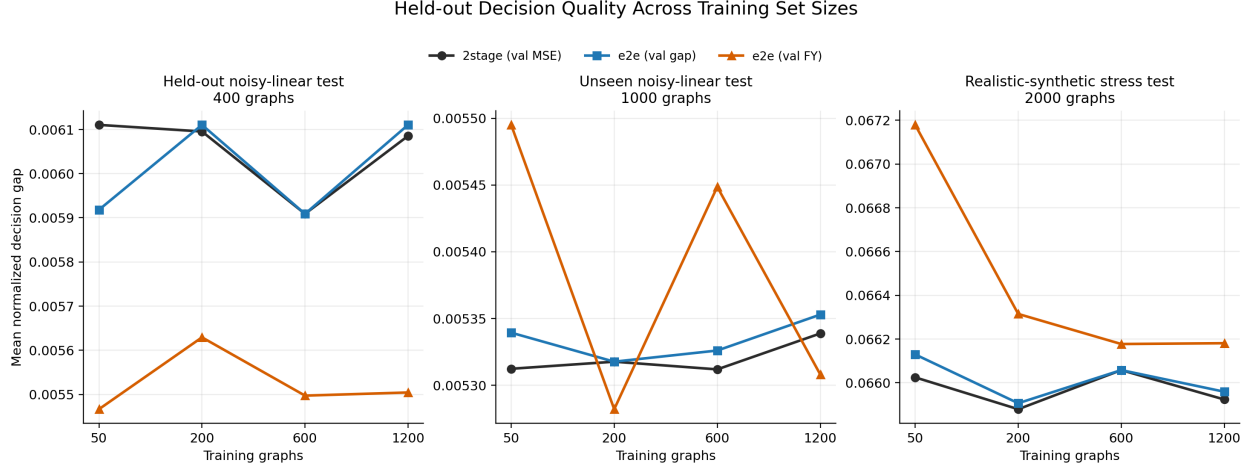


Figure 3: Mean normalized synthetic-label decision gap across training sizes and evaluation datasets. Lower is better.

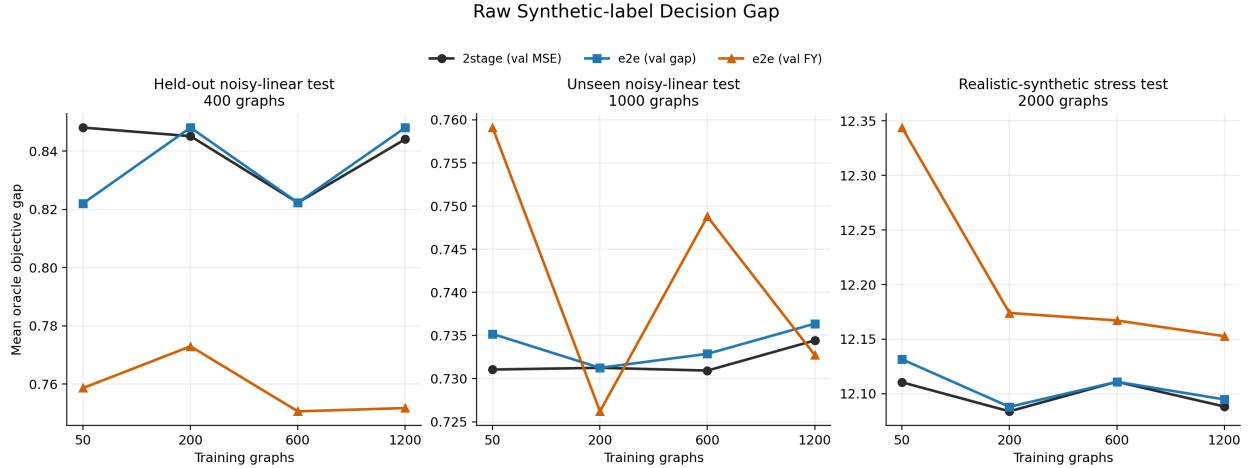


Figure 4: Mean raw oracle objective gap across datasets. Lower is better.

20.5 Selected parameter endpoints

How to explain this figure. This figure connects Step1b back to Step1a. The 2stage checkpoints cluster near the clean coefficient $[10, 5]$. The e2e checkpoint selected by validation decision gap may select smaller-scale parameters, while the e2e checkpoint selected by validation FY loss often selects a larger-scale parameter vector. This supports the interpretation that FY does not simply recover the reward coefficient; it moves according to perturbed decision stability.

20.6 Parameter trajectories

How to explain this figure. This figure shows the actual training paths. It helps explain why selected checkpoints differ: validation decision gap and validation FY loss can choose different epochs along the same FY trajectory. It also shows whether increasing training size stabilizes the trajectory.

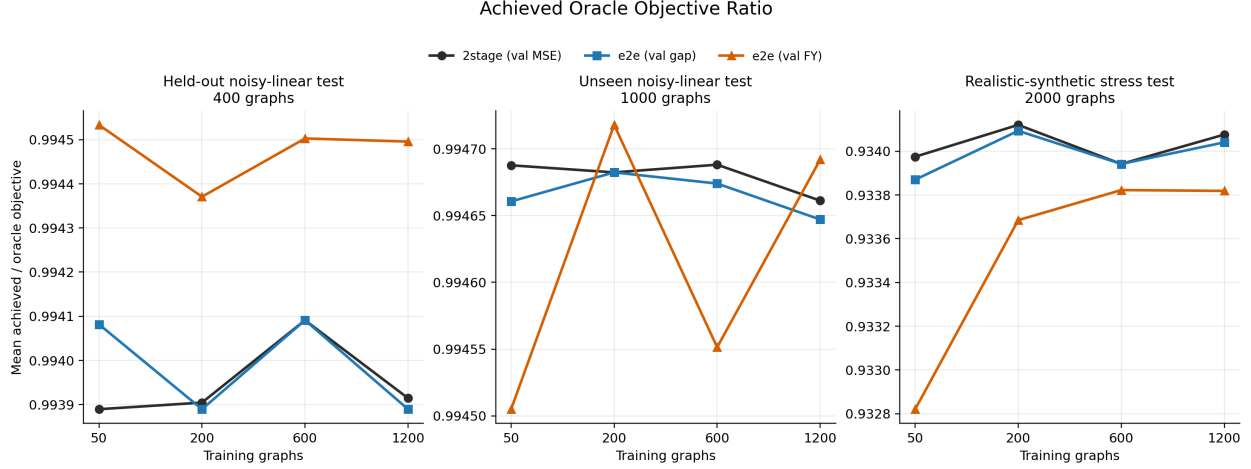


Figure 5: Mean achieved/oracle objective ratio. Higher is better.

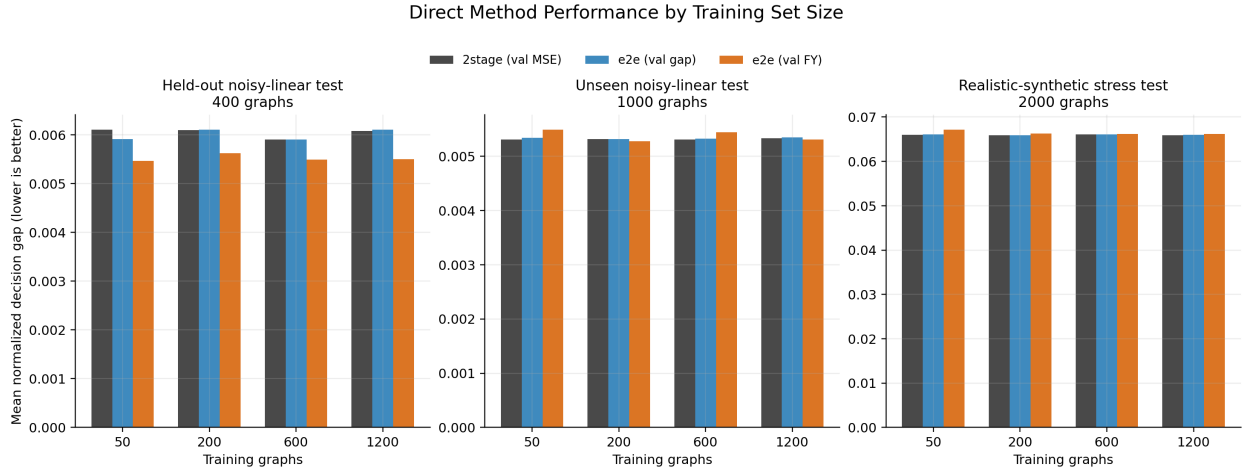


Figure 6: Direct method comparison by mean normalized decision gap. Lower values are better. Each training size shows three bars: 2stage selected by validation MSE, e2e selected by validation decision gap, and e2e selected by validation FY loss.

20.7 Training and validation curves

How to explain this figure. This combined figure keeps the three training diagnostics together. The first row is the 2stage reward-fitting sanity check: MSE should steadily fit the synthetic labels and validation MSE should track training MSE reasonably well. The second row shows whether the e2e FY surrogate objective is being optimized; the validation-FY-selected checkpoint is chosen from this curve. The third row is not the training objective, but it is the downstream decision metric used for validation-gap checkpoint selection.

20.8 Validation FY objective vs decision gap

How to explain this figure. This figure directly tests whether validation FY objective and validation decision gap are aligned. If the point minimizing FY objective is also near a low decision-gap point, then FY is useful not only for training but also for model selection. If the two markers

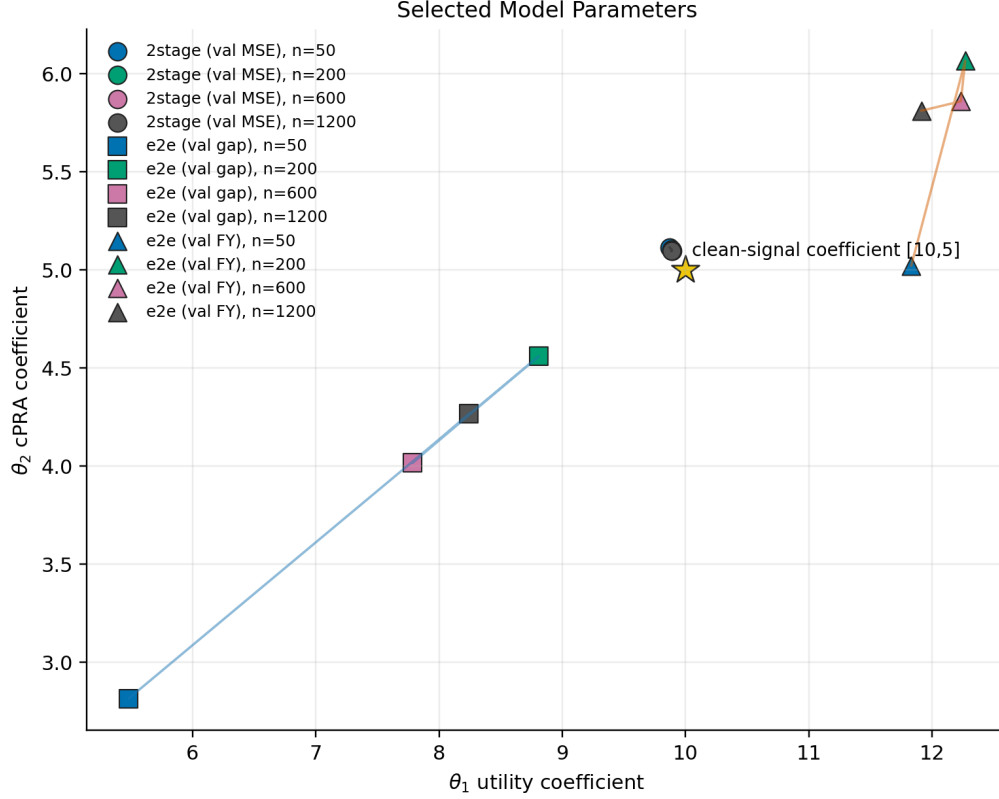


Figure 7: Selected model parameters in the two-dimensional linear-probe space.

are far apart, the FY objective may still provide useful gradients but may not be a reliable early-stopping criterion.

21 Main Interpretation

The current Step1b evidence is mixed but informative.

1. **On the original 400-graph held-out test split, FY-loss selection is promising.** For $n = 200$ and $n = 600$, the e2e checkpoint selected by validation FY loss improves over 2stage with positive paired improvement. This is the strongest evidence that the FY surrogate can be useful on the controlled noisy-linear benchmark.
2. **On the 1000-graph unseen noisy-linear dataset, the advantage is not robust.** The 2stage baseline remains stable, and e2e is tied or slightly worse in most cases. The only small positive e2e result is at $n = 200$ selected by validation FY loss, but its bootstrap interval crosses zero.
3. **Under the realistic-synthetic stress test, noisy-linear-trained e2e does not transfer better than 2stage.** This is expected to some extent because the model was trained on a simple noisy-linear label but evaluated under a more complex reward label.
4. **The two e2e selection rules answer different questions.** Validation decision gap asks whether the trajectory contains a good decision-quality point. Validation FY loss asks

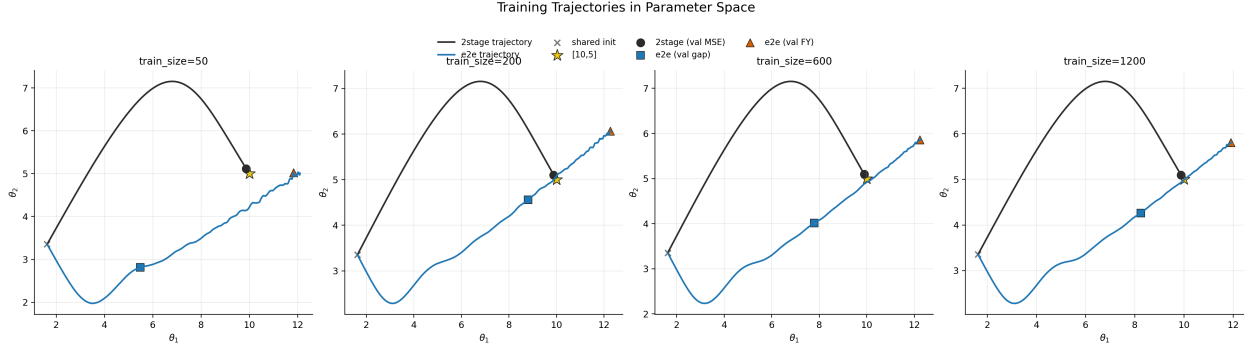


Figure 8: Training trajectories in (θ_1, θ_2) space for each training size.

whether the surrogate itself selects a good point. Reporting both is necessary.

5. **The current result suggests a next experimental step.** To make FY more competitive on unseen data, we may need better model capacity, stronger feature sets, warm-started FY fine-tuning, or tuning of ϵ , M , and checkpoint rules.

22 What We Can and Cannot Claim

Supported by current evidence. The FY training pipeline is now evaluated in a standard train/validation/test style. On the original held-out noisy-linear split, the FY-loss-selected checkpoint can outperform the two-stage MSE baseline for the completed $n = 200$ and $n = 600$ runs.

Not yet supported. The current evidence does not show a robust FY advantage on newly generated unseen noisy-linear graphs or under the realistic-synthetic label shift. Therefore, we should not claim that FY generally outperforms MSE on unseen KEP instances in the current two-feature probe setting.

Best summary sentence.

Step1b upgrades Step1a from an in-sample trajectory diagnostic to a held-out generalization study. The current results show that FY can select better decision-quality models on the original noisy-linear split, but this advantage is not yet robust on newly generated unseen graphs or under a shifted realistic-synthetic reward label.

23 Next Steps

The current report suggests the following next experimental steps:

1. Complete the $n = 1200$ formal run and add it to all tables/figures.
2. Add an MSE-warm-started FY fine-tuning method.
3. Repeat the formal runs over multiple subset/theta seeds.
4. Run an ϵ and M ablation.

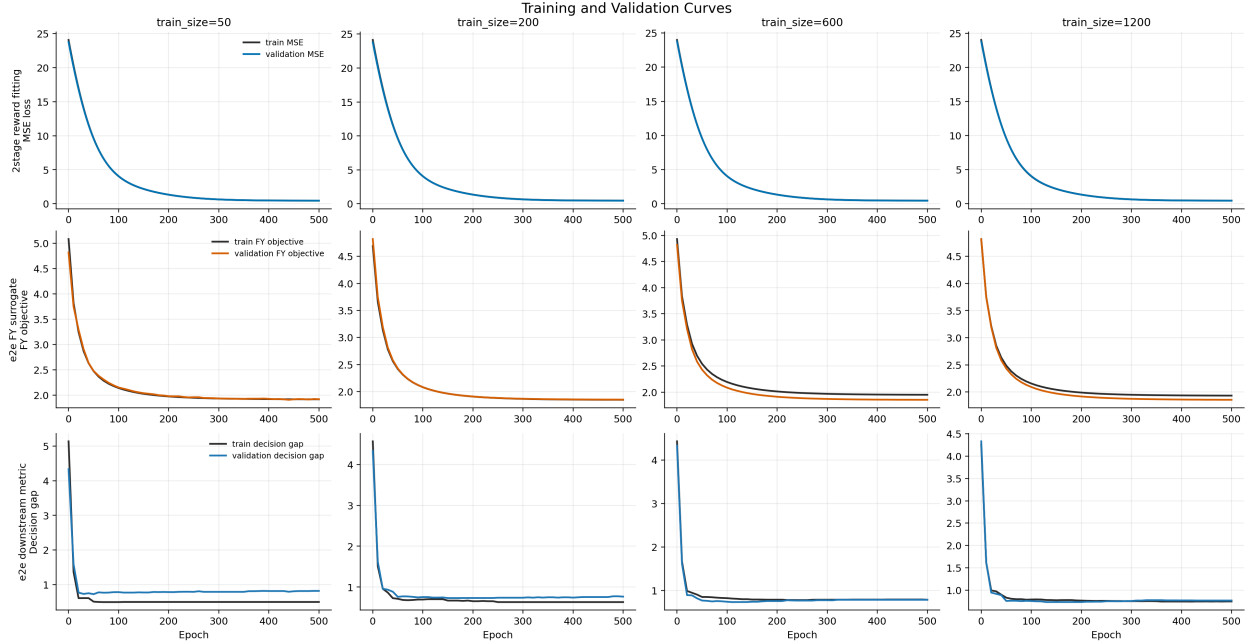


Figure 9: Training and validation curves for the three tracked quantities: 2stage MSE, e2e perturbed FY objective, and e2e synthetic-label decision gap. Rows correspond to metric families and columns correspond to training-set sizes. The 2stage MSE curves are evaluated at every epoch, while the e2e FY and decision-gap curves are evaluated every 10 epochs in the formal archive.

5. Test richer feature sets beyond utility and recipient cPRA.
6. Consider reporting both exact-zero-gap rate and nonzero-gap conditional performance, because many graphs produce identical decisions across methods.

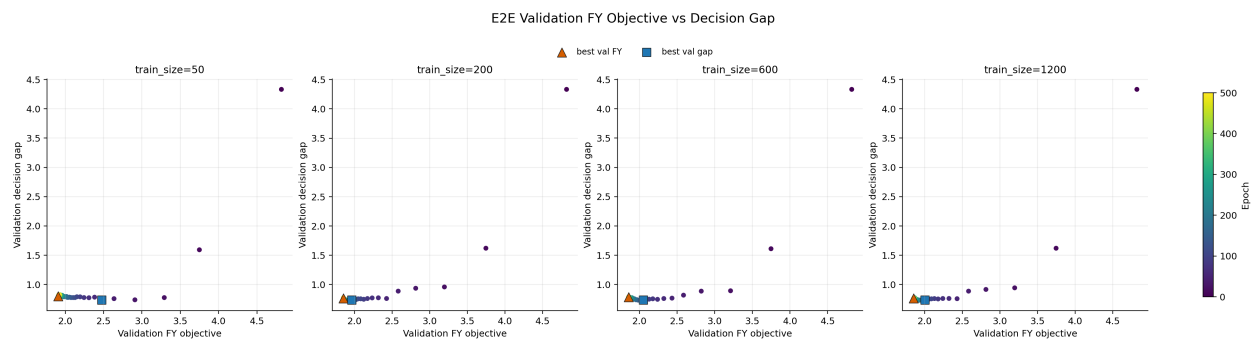


Figure 10: Relationship between validation FY objective and validation decision gap along the e2e trajectory.